ACM 104 Midterm

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1. a) True

Zero: $0^T = 0 = -0$

Closed $(+)$: $A^T = -A$, $B^T = -B \rightarrow (A+B)^T = A^T + B^T = -A - B = -(A+B)$

Closed $(\cdot)$: $A^T = -A \rightarrow (cA)^T = cA^T = c(-A) = -(cA)$

b) False

Counterexample: Let $v = (1,0)$, $w = (0,1)$, $u = (0,1) = 0 \cdot v + 1 \cdot w$

Then u is a linear combination of v and w, but $v = (1,0)$ cannot be written as a linear combination of u and w.

c) False

$\sin(x+1) = \sin x \cos 1 + \cos x \sin 1$

So $\sin(x+1)$ is a lin. comb. of $\sin x$ and $\cos x$ hence linear dependence.

d) False

$P_n = \{a_0 + a_1 x + \ldots + a_n x^n\}$ has basis $\{1, x, x^2, \ldots, x^n\}$ which contains $n+1$ elements, so $\dim(P_n) = n+1$.

e) True

By rank-nullity: $\operatorname{rank} A = n - \dim(\ker A)$; $\operatorname{rank} B = n - \dim(\ker B)$

Since $\ker A = \ker B$, $\dim(\ker A) = \dim(\ker B)$, so $\operatorname{rank} A = \operatorname{rank}(B)$

f) False

Counterexample: Let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

- $\ker A = \operatorname{span}\{(1,0)\}$ (vectors of form $(x,0)$)
- $\operatorname{im} A = \operatorname{span}\{(1,0)\}$ (Av gives vectors of form $(y,0)$)
- $\ker A \cap \operatorname{im} A = \operatorname{span}\{(1,0)\} \neq \{0\}$

g) False

Counterexample: Let $v = (1,0)$, $u = (0,1)$:

$\langle v, u \rangle = 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 1 = 1$

$\langle u, v \rangle = 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 = 0$

Since they are not equal, this is not symmetric and not an inner product.

h) True

For any $x \neq 0$: $x^T(A+B)x = x^TAx + x^TBx > 0 + 0 = 0$

Therefore $A+B$ is positive definite.

i) False

Counterexample: $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

For $x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $x^TAx = \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = -2 < 0$

So A is not positive semi-definite.

j) True

By reverse of Δ inequality: $\big| \|u\| - \|v\| \big| \leq \|u - v\|$

We get:

$$\big| \|u\| - \|-v\| \big| \leq \|u - (-v)\|$$

$$\big| \|u\| - \|v\| \big| \leq \|u + v\|$$

Therefore:

$$\|u\| - \|v\| \leq \|u+v\|$$

$$\|u\| \leq \|v\| + \|v+u\|$$

2. The numbers $a_1, \dots, a_n$ must not be the same to establish linear independence.

Proof

Assume $a_i = a_j$ for $i \neq j$, then $e^{a_i x} = e^{a_j x}$ for all x

This means:

$$1 \cdot e^{a_i x} + (-1) e^{a_j x} = 0$$

This yields linear dependence. Therefore $a_1, \dots, a_n$ must all be different to satisfy linear independance.

Sufficiency:

Suppose all a_i are unique. We can show that if

$$c_1 e^{a_1 x} + c_2 e^{a_2 x} + \dots + c_n e^{a_n x} = 0 \quad \forall x \in \mathbb{R}$$

then $c_1 = c_2 = \dots = c_n =$

We can differentiate the equation k times for $k \in [0, n-1]$, $k \in \mathbb{Z}$. This gives a linear system

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ a_1 & a_2 & \cdots & a_n \\ a_1^2 & a_2^2 & \cdots & a_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{n-1} & a_2^{n-1} & \cdots & a_n^{n-1} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

The coefficient matrix is a Vandermonde matrix with:

$$\det(V) = \prod_{1 \leq i < j \leq n} (a_j - a_i)$$

Since all a_i are different, each $(a_j - a_i) \neq 0$, so $\det(V) \neq 0$. Therefore the system has only the trivial solution and the functions are linearly dependent.

3. a) $p(-1)=0$ gives $p(-1)=a(-1)^2+b(-1)+c=a-b+c=0 \therefore c=b-a$

Any polynomial in V has the form:

$$p(x)=ax^2+bx+(b-a)=a(x^2-1)+b(x+1)$$

This shows $V=\text{span}\{x^2-1, x+1\}$

Check linear independence:

Let $a(x^2-1)+b(x+1)=0 \quad \forall x$

$$ax^2+bx+(-a+b)=0$$

Now we need $a=0, b=0, -a+b=0$ ✓

Thus $a=b=0$, so set is linearly independent.

$$\boxed{\begin{array}{l}\text{Basis: } \{x^2-1, x+1\} \\ \text{Dim: } 2\end{array}}$$

b) We can write symmetric matrices as $A=\begin{bmatrix} a & b \\ b & d \end{bmatrix}$ since $A^T=A$

$$A=a\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{E_1}+b\underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{E_2}+d\underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}}_{E_3}$$

Suppose $\alpha E_1+\beta E_2+\gamma E_3=0$

Then:

$$\begin{bmatrix} \alpha & \beta \\ \beta & \gamma \end{bmatrix}=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$\therefore \alpha=\beta=\gamma=0$

$$\boxed{\begin{array}{l}\text{Basis: } \left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right\} \\ \text{Dim: } 3\end{array}}$$

4) $A = \begin{bmatrix} 1 & -3 \\ 3 & -9 \end{bmatrix} \rightarrow$ rank 1 since cols are multiples

$\begin{bmatrix} 1 & -3 \\ 3 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x_1 = 3x_2, \; x_2 = t$

We can clearly see $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3t \\ t \end{bmatrix} = t \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

$\boxed{\text{Ker } A: \left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\}}$

$c_2 = \begin{bmatrix} -3 \\ -9 \end{bmatrix} = -3c_1 = -3 \begin{bmatrix} 1 \\ 3 \end{bmatrix} \rightarrow$ cols linearly dependent

$\boxed{\text{im } A: \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}}$

$\dim(\text{im } A) = 1 \rightarrow \dim(\text{coker } A) = 2 - 1 = 1$

$\begin{bmatrix} 3 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix} = 3 - 3 = 0$ $\qquad$ Coker A: $\left\{ \begin{bmatrix} 3 \\ -1 \end{bmatrix} + \text{im } A \right\}$

$\boxed{\text{coker } A: \left\{ \begin{bmatrix} 3 \\ -1 \end{bmatrix} \right\}}$

$\dim(\text{ker } A) = 1 \rightarrow \dim(\text{coim } A) = 2 - 1 = 1$

$\begin{bmatrix} 1 \\ -3 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 3 - 3 = 0$ $\qquad$ coim A: $\left\{ \begin{bmatrix} 1 \\ -3 \end{bmatrix} + \text{ker } A \right\}$

$\boxed{\text{coim } A: \left\{ \begin{bmatrix} 1 \\ -3 \end{bmatrix} \right\}}$

Check: By rank-nullity:

$\dim(\text{ker } A) = 1, \dim(\text{im } A) = 1 \rightarrow 1 + 1 = 2 = \dim(\mathbb{R}^2)$ $\checkmark$

5.

a) $\|v+w\|^2 = \langle v+w, v+w\rangle = \langle v, v\rangle + 2\langle v, w\rangle + \langle w, w\rangle = \|v\|^2 + \|w\|^2$

only because $2\langle v, w\rangle = 0$ b/c $v \perp w$

b) $\|v+w\|^2 = \|v\|^2 + 2\langle v, w\rangle + \|w\|^2$

$\|v-w\|^2 = \|v\|^2 - 2\langle v, w\rangle + \|w\|^2$

$\|v+w\|^2 + \|v-w\|^2 = 2\|v\|^2 + 2\|w\|^2$

c) Let $U_v = \frac{v}{\|v\|}$ and $U_w = \frac{w}{\|w\|}$

$$\|U_v - U_w\|^2 = \|U_v\|^2 - 2\langle U_v, U_w\rangle + \|U_w\|^2$$

$$= 1 - 2\langle U_v, U_w\rangle + 1 = 2 - 2\langle U_v, U_w\rangle$$

$$= 2 - 2\left\langle \frac{v}{\|v\|}, \frac{w}{\|w\|}\right\rangle$$

$$= 2 - 2\,\frac{\langle v, w\rangle}{\|v\|\,\|w\|}$$

$$\frac{1}{2}\|U_v - U_w\|^2 = 1 - \frac{\langle v, w\rangle}{\|v\|\,\|w\|}$$

$$\langle v, w\rangle = \|v\|\,\|w\|\left(1 - \frac{1}{2}\left\|\frac{v}{\|v\|} - \frac{w}{\|w\|}\right\|^2\right)$$

6. $\nabla f = \langle -1-2x-z, -1-2y, 1-2z-x \rangle = 0$

$$\left.\begin{aligned} 2x+z &= -1 \\ 2y &= -1 \rightarrow y = -\tfrac{1}{2} \\ x+2z &= 1 \end{aligned}\right\} \longrightarrow x = -1,\ z = 1$$

$$f\left(-1, -\tfrac{1}{2}, -1\right) = 1-(-1)-\left(-\tfrac{1}{2}\right) - 1^2 - (-1)^2 - \left(-\tfrac{1}{2}\right)^2 - (-1\cdot 1)$$
$$= 1.25$$

$$H = \begin{bmatrix} -2 & 0 & -1 \\ 0 & -2 & 0 \\ -1 & 0 & -2 \end{bmatrix} \longrightarrow \begin{aligned} H_1 &= -2 < 0 \ \checkmark \\ H_2 &= \begin{vmatrix} -2 & 0 \\ 0 & -2 \end{vmatrix} = 4 > 0 \ \checkmark \\ H_3 &= \det(H) = -2(4) - (-1)(0\cdot 0 - (-2)(-1)) \\ &= -8-2 = -6 < 0 \ \checkmark \end{aligned}$$

Because of checks and alternating signs, H is negative definite so local max.

7.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \\ -1 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} x \\ y \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}$$

$$A^T A x = A^T b$$

$$A^T = \begin{bmatrix} 1 & 3 & -1 \\ 2 & -1 & 2 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 3 & -1 \\ 2 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 11 & -3 \\ -3 & 9 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 3 & -1 \\ 2 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} -3 \\ 10 \end{bmatrix}$$

$$\begin{bmatrix} 11 & -3 \\ -3 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -3 \\ 10 \end{bmatrix}$$

$$\begin{aligned} 11x - 3y &= -3 \\ -3x + 9y &= 10 \end{aligned} \longrightarrow \begin{aligned} 33x - 9y &= -9 \\ -3x + 9y &= 10 \end{aligned}$$

$$30x = 1 \therefore \boxed{x = \frac{1}{30}, \; y = \frac{101}{90}}$$

8. a) Show $x^T M x \geq 0 \quad \forall x$

$$x^T(A^TA)x = (x^TA^T)(Ax) = (Ax)^T(Ax) = ||Ax||^2 \geq 0$$

$\therefore$ A^TA is positive semi-definite.

b) i) Show $\ker(A) \subseteq \ker(A^TA)$

If $x \in \ker(A)$, then $Ax = 0$

$\therefore A^TAx = A^T(Ax) = A^T(0) = 0$

So $x \in \ker(A^TA)$, so $\ker(A) \subseteq \ker(A^TA)$

ii) Show $\ker(A^TA) \subseteq \ker(A)$

If $x \in \ker(A^TA)$, then $A^TAx = 0$

$\therefore \quad x^T(A^TAx) = x^T \cdot 0 = 0$

$x^TA^TAx = 0$

$(Ax)^T(Ax) = 0$

$||Ax||^2 = 0 \qquad \therefore Ax = 0$

So $x \in \ker(A)$, so $\ker(A^TA) \subseteq \ker(A)$

Therefore: $\ker(A^TA) = \ker(A)$

c) We can do rank-nullity from $\mathbb{R}^m \to \mathbb{R}^n$

$$\text{rank}(A) + \dim(\ker(A)) = n \qquad (1)$$

For $A^TA : \mathbb{R}^n \to \mathbb{R}^m$:

$$\text{rank}(A^TA) + \dim(\ker(A^TA)) = n \qquad (2)$$

Since $\ker(A^TA) = \ker(A)$ from (b):

$$\dim(\ker(A^TA)) = \dim(\ker(A))$$

Setting (1) = (2)

$$\text{rank}(A^TA) + \dim(\ker(A)) = \text{rank}(A) + \dim(\ker(A))$$

Therefore:

$$\boxed{\text{rank}(A^TA) = \text{rank}(A)}$$